AI-based agricultural market intelligence research

"Analyze research activity on AI-based agricultural market intelligence, crop price forecasting, multi-farmer crop planning, and crop oversupply prediction (2020-2026). Using verifiable Scopus or Google Scholar data, provide the total number of papers retrieved, total citations, and average citations per paper. Explain what these metrics indicate about research trends and academic interest. Include the search keywords, database, search date, and sources." *Generated by LeapSpace*

AI-Based Agricultural Market Intelligence and Crop Forecasting at a Glance

Key Findings Table

Topic	Main finding	Apparent research activity	What the evidence supports	Citations
AI-based agricultural market intelligence	Research emphasizes decision-support components such as crop recommendation, yield prediction, disease detection, and price prediction, but verified end-to-end systems integrating agronomy and market intelligence remain rare	Moderate	The area is active, but integration across recommendation, forecasting, and advisory is still limited	[1] [2] [3] [4]
Crop price forecasting	This is the densest and most mature subfield in the supplied evidence, with frequent use of LSTM, GRU, CNN-LSTM, ARIMA, SARIMA, ensembles, and decomposition-based hybrids	High	Publication attention appears strongest here, especially for predictive accuracy benchmarking	[5] [6] [7] [8] [9] [10] [11] [12]
Multi-farmer crop planning	Most studies focus on single-site or single-farmer recommendation; explicit cooperative or multi-farmer optimization is weakly represented	Low to moderate	This is a clear research gap rather than a mature stream	[13] [14] [15] [16] [17] [18] [19]
Crop oversupply prediction	Oversupply is usually modeled indirectly through demand forecasting, inventory balancing, early warning, or price prediction rather than direct surplus forecasting	Low	The subfield is emerging and conceptually fragmented	[20] [21] [22] [23]
Methodological trend	Hybrid, ensemble, and multivariate approaches are favored over single-model univariate baselines	High	The field is moving toward richer exogenous-data modeling	[6] [7] [8] [9] [11] [12]
Deployment reality	Reviews describe movement toward operational dashboards, mobile advisory, and real-time systems, but adoption barriers remain substantial	Medium	Practical deployment lags algorithmic progress	[24] [25] [26] [27] [28] [29]

Direct Answer

The supplied evidence supports a clear qualitative conclusion: from 2020-2026, research activity is strongest in **AI-based crop price forecasting**, moderate in **agricultural decision-support/market-intelligence components**, and much thinner in **explicit multi-farmer crop planning** and **direct crop oversupply prediction** [1] [3] [5] [10] [15] [21].

However, the requested **verifiable bibliometric outputs—total papers retrieved, total citations, and average citations per paper from Scopus or Google Scholar—cannot be reliably computed from the supplied evidence alone** because the input provides topical findings and paper-level learnings, but not raw database hit counts, citation-export totals, or an executed search log with reproducible database/date metadata. The evidence therefore supports a **research-activity analysis**, but not a defensible bibliometric count table.

What the available evidence does show is that academic interest is concentrated where tasks are well-defined and data are easier to obtain—especially price forecasting—while coordinated market problems such as oversupply prevention and multi-farmer planning remain less standardized and less developed [7] [10] [17] [23] [30].

Confidence level: High for the qualitative ranking of subfields, because multiple topic clusters converge on the same pattern [3] [10] [15] [20].

Study Scope

- **Period covered:** 2020-2026, as defined by the query and reflected in the retrieved topic summaries.
- **Topical scope:** AI-based agricultural market intelligence, crop price forecasting, crop planning/allocation, and oversupply or demand-supply imbalance prediction [1] [10] [14] [21].
- **Methods covered:** machine learning, deep learning, time-series forecasting, hybrid decomposition models, optimization/metaheuristics, and decision-support systems [9] [11] [14] [31] [32].
- **Evidence type:** synthesized learnings from retrieved studies and reviews; not a raw Scopus or Google Scholar export.

Assumptions & Limitations

- The evidence base is sufficient for **qualitative synthesis**, but not for **verifiable bibliometric counting**, because raw retrieval counts, citation totals, database result pages, and search dates are not provided.
- Search strings are partially recoverable from the topic iterations, but **database identity and executed search date are not fully documented** in the supplied material.
- Some conclusions about maturity are inferential, based on repeated thematic patterns across reviews and primary studies rather than direct bibliometric tallies [10] [23] [24] [33].

Suggested Further Research

- Build a reproducible Scopus/Google Scholar search log with exact query strings, search date, inclusion rules, and exported citation counts.
- Develop integrated systems combining crop recommendation, price forecasting, and oversupply early warning in one pipeline [3] [4] [21].
- Expand from single-farm recommendation to cooperative and multi-farmer optimization models [15] [17] [19].
- Standardize operational definitions of “oversupply” and “surplus risk” for comparative modeling [20] [21] [23].
- Test explainable and deployable models with real-world advisory workflows, not only benchmark accuracy [25] [28].

Full Report

Foundational Context

The four requested areas are related but not identical. **Agricultural market intelligence** concerns decision support for production, marketing, timing, and sometimes logistics; **crop price forecasting** is a narrower predictive task focused on market prices; **crop planning** concerns what to plant and where, often under resource or risk constraints; and **oversupply prediction** concerns imbalance between expected production and downstream demand, storage, or offtake capacity [5] [13] [21] [34] [35].

This distinction matters because publication volume is not evenly distributed. Price forecasting is easier to benchmark with standard error metrics such as MAE and RMSE, whereas multi-farmer planning and oversupply forecasting require more complex definitions, richer data, and broader coordination assumptions [17] [30] [36] [37]. Taken together, these findings suggest that methodological maturity rises as the task becomes more narrowly defined and data-amenable [10] [23] [33].

Confidence level: High - this framing is directly supported across all four topic clusters [1] [5] [14] [21].

Search Strategy and Reproducibility Status

Recoverable search keywords from the supplied evidence

The input includes the following topic-cluster search formulations:

- **Agricultural market intelligence / DSS:** `(AI OR machine learning OR deep learning) AND (agricultural market intelligence OR market intelligence OR decision support system OR decision-support) AND (crop OR agriculture)` [topic iteration text in input].

- **Crop price forecasting:** `(crop price forecasting OR agricultural price prediction OR market price prediction) AND (machine learning OR deep learning OR time series OR econometric)` [topic iteration text in input].
- **Crop planning/allocation:** `(multi-farmer crop planning OR crop planning OR portfolio optimization OR land allocation OR crop allocation) AND (AI OR machine learning OR optimization OR decision support)` [topic iteration text in input].
- **Oversupply / imbalance:** `(crop oversupply OR surplus prediction OR demand-supply imbalance OR supply-demand imbalance OR market surplus) AND (forecasting OR prediction OR machine learning OR AI)` [topic iteration text in input].

Database, search date, and source-verification status

The query requests **Scopus or Google Scholar** counts, but the supplied evidence does **not** provide an executed database log, a search date, or raw result-count screenshots/exports. Accordingly:

Reproducibility item	Status from supplied evidence
Exact keywords	Partially available via topic-cluster search strings
Database used	Not verifiable from supplied evidence
Search date	Not reported
Total papers retrieved	Not verifiable
Total citations	Not verifiable
Average citations per paper	Not computable
Source records	Available only as citation-ID-linked synthesized findings

This means the present report can audit **research themes and relative activity**, but cannot defensibly claim numerical bibliometric totals.

Confidence level: High - this is a statement about what is and is not contained in the supplied evidence.

Bibliometric Results: What Can and Cannot Be Reported

Requested metrics

The user requested:

1. total number of papers retrieved,

2. total citations,
3. average citations per paper,
4. search keywords, database, search date, and sources.

Only the **search keywords** are recoverable with reasonable confidence from the supplied inputs. The remaining bibliometric items are not directly available.

What the absence of countable metadata implies

Because the input does not include raw Scopus or Google Scholar exports, any numerical total for papers or citations would be a reconstruction rather than a verifiable bibliometric result. That would violate the user's requirement for **verifiable** database-based metrics. The strongest defensible answer is therefore methodological honesty: **the supplied evidence does not support those counts**.

Still, the topical synthesis is consistent enough to indicate relative research concentration:

- **Highest activity:** crop price forecasting [5] [10].
- **Moderate activity:** AI-enabled decision support and market-intelligence components [1] [24].
- **Lower activity:** multi-farmer/cooperative crop planning [15] [17].
- **Lowest explicit activity:** direct oversupply prediction [20] [23].

Confidence level: High for relative activity; **Low** for any unreported numerical bibliometric total, which should not be inferred.

Research Themes and Evidence Patterns

1) AI-driven agricultural market intelligence and decision support

The evidence shows a broad decision-support literature in agriculture, spanning monitoring, yield prediction, pest detection, irrigation, crop recommendation, and some price prediction [29] [38] [39] [40] [41]. Reviews characterize the field as moving from concept toward execution, with increasing integration of IoT, sensing, robotics, and predictive analytics [24] [26] [42].

At the application level, some systems combine forecasting with farmer-facing interfaces. Examples include dashboards using historical commodity prices plus external economic and environmental factors, and systems using local mandi, wholesale, weather, and demand data for sale timing and market selection [1] [2] [36]. But the evidence also states that no retrieved full-text study clearly integrates crop or input recommendation with market-price or demand forecasting in a single end-to-end pipeline [3] [4].

This creates an important distinction: the field contains many **components** of market intelligence, but relatively few truly integrated **market-intelligence systems** [1] [3].

Confidence level: High - multiple reviews and study-level summaries support this pattern [2] [4] [24].

2) Crop price forecasting methods

Crop price forecasting is the most developed stream in the supplied evidence. Common methods include ARIMA, SARIMA, LSTM, GRU, CNN-LSTM, Bi-LSTM, NNAR, stacked ensembles, and decomposition-based hybrids [5] [9] [11] [12] [28] [43] [44] [45].

A recurring pattern is that **hybrid and multivariate models often outperform simpler baselines**. Examples include ARIMA-LSTM with random-forest lag selection for volatile pulses, VMD-SGMD-LSTM for futures, and VMD-LSTM outperforming several decomposition-LSTM alternatives [9] [11] [12]. Reviews explicitly identify combined models as the likely future direction, especially when structured and unstructured data can be integrated [10].

However, AI superiority is not uniform. Traditional models outperformed deep learning in a barley case, and a double moving average beat NNAR in one wheat comparison [46] [47]. This means the literature favors AI and hybrids overall, but not as a universal rule [6] [46] [47] [48] [49].

Operationally, these models are positioned as tools for planting, harvesting, trading, and risk management decisions [5] [45] [50] [51]. Some studies go further and report simulated decision gains, such as higher profit and lower variance from integrated yield-price systems [52].

Confidence level: High - this is the best-supported and most internally consistent subfield in the supplied evidence [10] [28] [53].

3) Multi-farmer crop planning and allocation optimization

The crop-planning literature is active, but it is mostly centered on **single-farm or single-site recommendation**, not true multi-farmer coordination [18] [35] [54] [55] [56]. Studies use soil, climate, weather, and historical yield data for crop recommendation and planning, while reviews emphasize broader AI-enabled precision management [33] [57] [58].

There are relevant optimization-oriented exceptions. One land-allocation study used AMIS plus a mathematical model to optimize planting and harvesting under yield, price fluctuation, and incompatibility constraints to maximize annual income [13]. Another combined metaheuristics with ML yield prediction for large-scale crop allocation [14]. These indicate that optimization work exists, but it is not yet strongly centered on explicit cooperative or multi-farmer coordination [15] [17].

The strongest direct conclusion is that **multi-farmer and cooperative planning remains underrepresented** relative to individualized recommendation systems [15] [16] [17] [19].

Confidence level: High - the gap is repeatedly stated across both study and review summaries [15] [17] [18].

4) Oversupply and demand-supply imbalance prediction

The oversupply literature is comparatively immature. AI and ML are used to forecast prices, demand, production, and supply-chain inventory conditions, but crop oversupply is usually handled indirectly [2] [5] [32] [59] [60] [61]. The most relevant formulations define imbalance using output, consumption, and price, or forecast shortages/surpluses from production, climate, and market data [20] [21].

Even here, the field lacks a standardized surplus concept. Many studies infer oversupply risk from demand forecasting or inventory balancing rather than directly predicting commodity-specific agricultural surplus [22] [62] [63] [64]. Reviews emphasize data timeliness, quality, and policy translation as major constraints [23] [30] [65].

Taken together, these findings suggest that oversupply prediction is an **emerging frontier** rather than a mature, benchmark-driven field [21] [23].

Confidence level: High - the indirect nature of oversupply modeling is consistently described [20] [21] [22].

Interpretation of Research Trends and Academic Interest

If full Scopus or Google Scholar counts were available, publication totals and citations would indicate scale, visibility, and consolidation. In their absence, the thematic evidence still supports several trend interpretations.

First, **academic interest is strongest in tasks with measurable targets and accessible time-series data**, especially crop price forecasting [5] [10]. Second, **hybrid and multivariate modeling has become the dominant methodological direction**, indicating a field moving beyond simple univariate extrapolation toward richer market-context prediction [7] [8] [11]. Third, **deployment interest is rising**, with dashboards, mobile advisory, and real-time systems appearing more often, though adoption barriers remain significant [25] [28] [66].

Conversely, relatively sparse work on multi-farmer coordination and direct oversupply prediction suggests that these areas have **high practical importance but lower methodological standardization and weaker data infrastructure** [17] [23] [30]. This pattern usually characterizes an emerging or fragmented subfield rather than a saturated one.

Inferred: Taken together, these findings imply a field with strong momentum but uneven maturity: mature enough to produce competitive forecasting benchmarks, yet still early in integrating forecasting with cooperative planning and market-balancing interventions. **Confidence level: Medium** - this is a synthesis across topic clusters rather than a directly measured bibliometric result [10] [23] [24] [33].

Practical Implications

For researchers, the main opportunity is integration: linking **price forecasting**, **crop allocation**, and **oversupply early warning** in a single decision framework [3] [14] [21]. For practitioners, current evidence suggests that deployable value is most immediate in **market-price forecasting and advisory dashboards**, not yet in cooperative-scale optimization [2] [15] [28]. For policymakers and cooperatives, the evidence gap itself is important: robust systems for coordinated planting and surplus prevention are still underdeveloped [17] [23].

Conclusion

A defensible quantitative bibliometric summary cannot be produced from the supplied evidence because it lacks raw Scopus or Google Scholar retrieval counts, citation totals, and search-date metadata. What can be concluded with high confidence is that, in 2020-2026, **crop price forecasting** is the most active and methodologically mature AI subfield in this domain; **agricultural market intelligence** is growing but still fragmented across components; **multi-farmer crop planning** is underdeveloped; and **direct oversupply prediction** remains an emerging area usually modeled only indirectly [3] [10] [15] [21].

Overall conclusion confidence: High for thematic and trend interpretation; **Low** for any requested bibliometric total not explicitly present in the supplied evidence.

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